

SIMATS ENGINEERING
CO2_ASSESSMENT TOOL3_Application Based Questions

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1522

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COURSE OUTCOME COVERED

CO2: Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

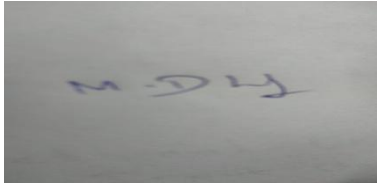
Total Marks: 25

Code of Conduct

I DHILIBAN M/192421166 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Assessment Tasks

Application Based Questions

1. A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.
2. An organization wants to store large volumes of backup data securely in the cloud. Discuss how cloud storage solutions ensure durability and data redundancy. How can encryption and access control protect sensitive backup data? Explain the cost advantages compared to traditional storage systems. Provide an example of a disaster recovery use case.

3. A mobile app company plans to deploy its application globally. Explain how content delivery networks (CDNs) improve performance and reduce latency. How do edge servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.
4. A financial institution requires strict compliance and data security in cloud usage. Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions.
5. A development team wants to adopt containerization for application deployment. Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments. Provide an example of a microservices-based application.

Application Based Questions Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Relevance to Application	25%	Response directly addresses the given use case with strong relevance	Mostly relevant response	Partially relevant	Weak relevance
Use of Cloud Concepts	25%	Applies appropriate concepts accurately	Mostly accurate application	Basic application only	Incorrect concept usage
Practical Insight	20%	Shows strong real-world understanding	Shows reasonable practical understanding	Limited practical insight	No practical insight
Completeness	15%	Response covers all required aspects	Covers most aspects	Covers only basic aspects	Incomplete response
Clarity	15%	Clear and concise answer	Generally clear	Some confusion in expression	Poor clarity

1. A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.

Handling Demand:

- Auto-scaling automatically adds/removes compute instances based on traffic.
- Load balancers distribute requests evenly across servers, preventing overload.

Reduced Downtime & Improved UX:

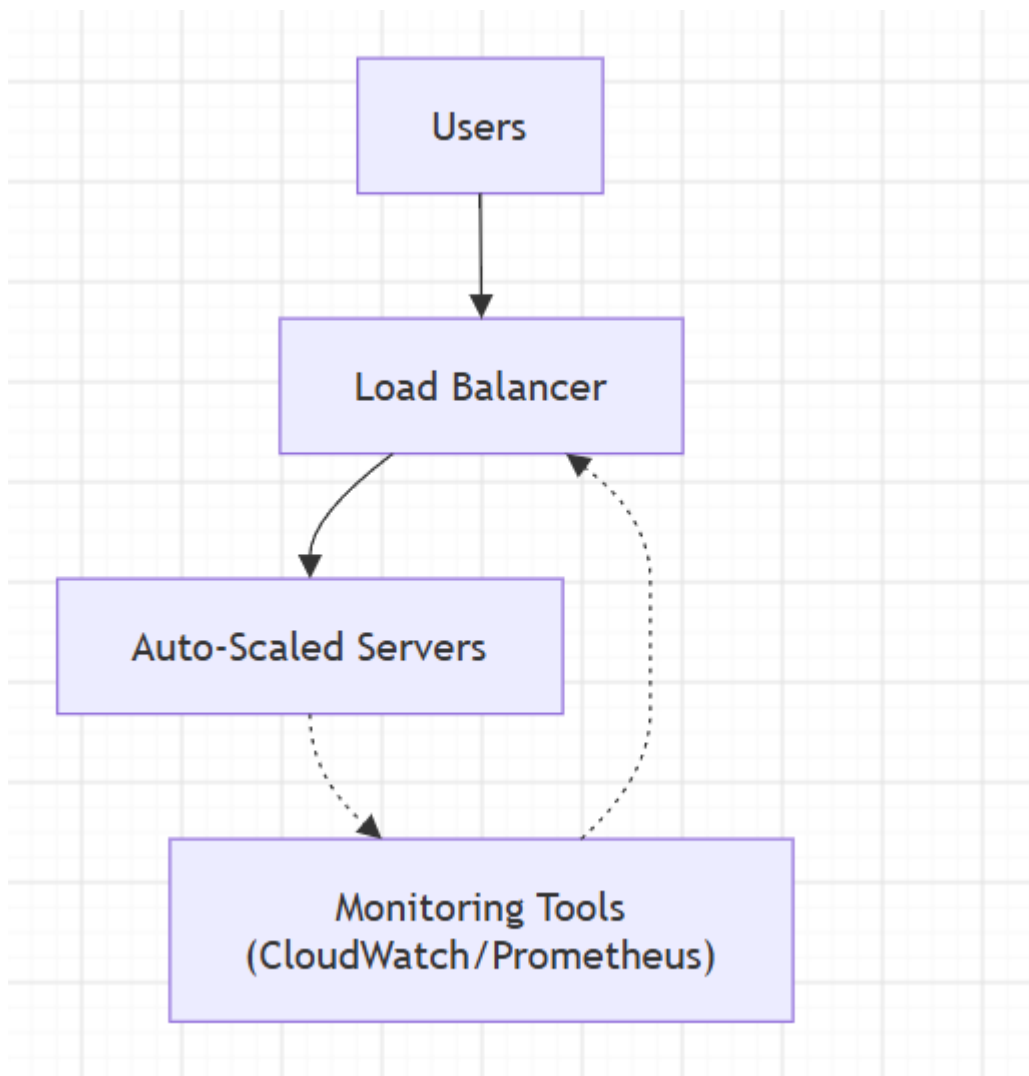
- Prevents single-server crashes during spikes.
- Ensures consistent response times and smooth checkout experience.

Monitoring Tools:

- CloudWatch, Datadog, or Prometheus track CPU, memory, and traffic.
- Alerts trigger scaling actions before bottlenecks occur.

Real-World Example:

- Amazon during Black Friday uses AWS Auto Scaling + Elastic Load Balancing to handle millions of concurrent shoppers.



2. An organization wants to store large volumes of backup data securely in the cloud. Discuss how cloud storage solutions ensure durability and data redundancy. How can encryption and access control protect sensitive backup data? Explain the cost advantages compared to traditional storage systems. Provide an example of a disaster recovery use case.

- **Durability & Redundancy:**

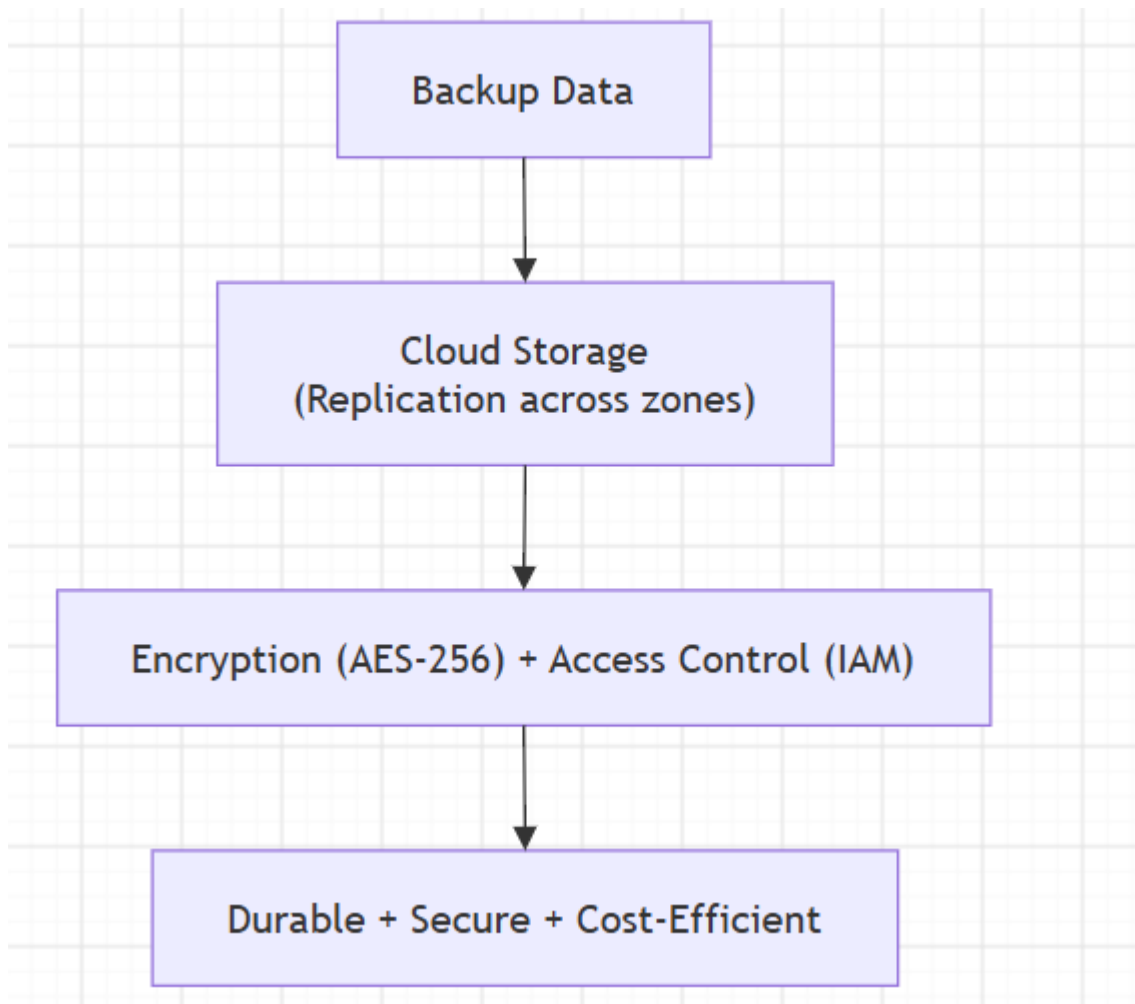
- Cloud storage (AWS S3, Azure Blob) replicates data across multiple regions/zones.
- Ensures “11 nines” durability (99.999999999%).

- **Encryption & Access Control:**

- AES-256 encryption at rest, TLS in transit.
- IAM policies and role-based access prevent unauthorized access.

- **Cost Advantages:**

- Pay-as-you-go vs. upfront hardware costs.
- Tiered storage (hot, cold, archival) optimizes expenses.
- **Disaster Recovery Example:**
 - A hospital backs up patient records to cloud; in case of ransomware attack, restores clean copies from cloud backups.



3. A mobile app company plans to deploy its application globally. Explain how content delivery networks (CDNs) improve performance and reduce latency. How do edge servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.

Performance & Latency:

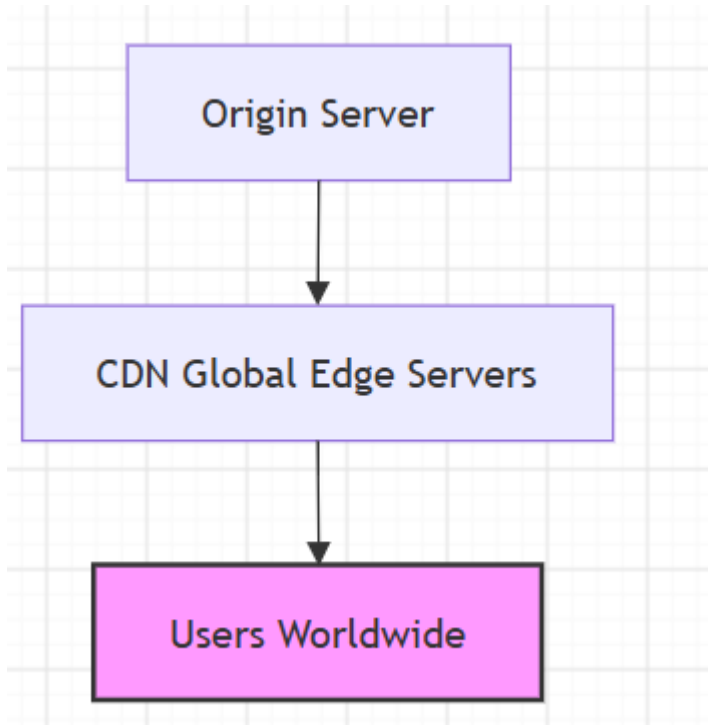
- CDNs cache content closer to users, reducing round-trip time.
- Edge servers serve static/dynamic content locally.
- **Enhanced UX:**
 - Faster load times, reduced buffering, smoother gameplay/streaming.

- **Cloud Provider Support:**

- AWS CloudFront, Azure CDN, Google Cloud CDN provide global PoPs.

- **Example:**

- Netflix uses CDNs + edge caching to deliver HD/4K streams worldwide with minimal lag.



4.A financial institution requires strict compliance and data security in cloud usage. Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions.

- **Regulatory Compliance:**

- Cloud providers offer certifications (PCI DSS, HIPAA, GDPR).
- Built-in auditing and compliance dashboards.

- **Security Tools:**

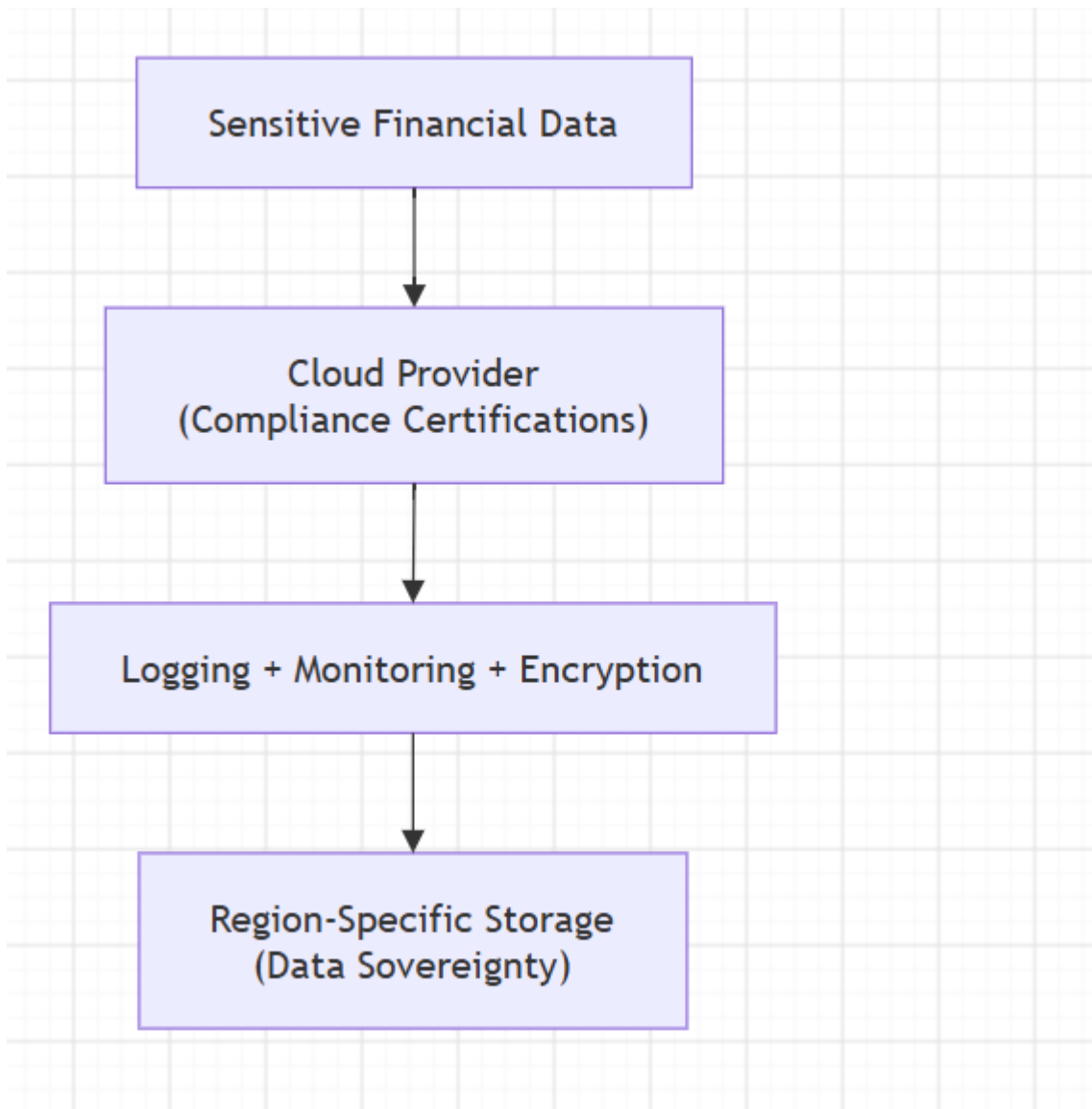
- Logging (CloudTrail), monitoring (SIEM), encryption (KMS/HSM).

- **Data Sovereignty:**

- Choose region-specific data centers to meet local laws.

- **Example:**

- A bank processing credit card transactions ensures logs are immutable, data encrypted, and stored in country-specific regions to comply with RBI/PCI DSS.



5.A development team wants to adopt containerization for application deployment. Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments. Provide an example of a microservices-based application.

- **Containers vs. VMs:**
 - Containers share OS kernel → lightweight, faster startup, lower resource usage.
 - VMs emulate hardware → heavier, slower, more resource-intensive.
- **Orchestration (Kubernetes):**
 - Automates deployment, scaling, service discovery, and self-healing.

- Simplifies rolling updates and fault tolerance.
- **Portability:**
 - Same container image runs on AWS, Azure, GCP, or on-prem.
- **Example:**
 - An e-commerce microservices app: separate containers for product catalog, payment, and recommendation engine, orchestrated via Kubernetes.

